

SOC Log Analysis & Incident Detection

Project Report

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Project Type: SOC / Cybersecurity Log Analysis

Tools Used: Linux Authentication Logs, VS Code/Notepad, Basic Log Filtering Techniques

1. Introduction

Security Operations Centers (SOCs) continuously monitor system and network activity to identify suspicious behavior, detect threats, and investigate security incidents. One of the most important responsibilities of a SOC analyst is analyzing authentication logs to identify failed login attempts, brute-force attacks, unauthorized access attempts, and suspicious user activity.

This project focuses on analyzing Linux authentication logs to detect suspicious authentication behavior and investigate possible security incidents. Through log inspection, filtering, and event correlation, multiple indicators of brute-force activity and unauthorized access attempts were identified.

2. Objective

The objectives of this project were:

- Analyze Linux authentication logs
- Identify failed login attempts
- Detect suspicious IP addresses
- Investigate repeated authentication failures
- Detect brute-force attack patterns
- Apply basic log filtering techniques
- Understand SOC-style investigation workflows
- Explore practical security monitoring concepts

3. Tools Used

Tool	Purpose
Linux Authentication Logs (auth.log)	Security event analysis
VS Code / Notepad	Log inspection and filtering
Search / Filtering Techniques	Event investigation

4. Authentication Log Overview

Linux authentication logs record security-related events associated with user authentication and privilege usage.

The analyzed logs included:

- Failed password attempts
- Successful user logins
- Sudo privilege usage
- SSH authentication activity
- Invalid username login attempts

Authentication logs are highly important in SOC environments because they help identify:

- Unauthorized access attempts
- Credential guessing attacks
- Privileged account targeting
- Brute-force behavior
- Suspicious login activity

5. Failed Login Analysis

Authentication logs were analyzed to identify failed login attempts and suspicious authentication activity.

Multiple failed password attempts were detected from external IP addresses targeting privileged accounts such as root and invalid users including admin and test.

Observations

- Repeated failed login attempts were observed
- Invalid usernames such as admin and test were targeted
- Multiple authentication failures occurred within short time intervals
- Failed root login attempts were identified

Repeated failures occurring within short time intervals may indicate:

- Brute-force attacks
- Credential guessing activity
- Unauthorized access attempts

Screenshot

```
Feb 24 10:15:32 kali sshd[1024]: Failed password for invalid user admin from 192.168.1.50 port 53422 ssh2
Feb 24 10:15:35 kali sshd[1024]: Failed password for invalid user admin from 192.168.1.50 port 53422 ssh2
Feb 24 10:15:38 kali sshd[1024]: Failed password for invalid user admin from 192.168.1.50 port 53422 ssh2
Feb 24 10:16:01 kali sshd[1050]: Accepted password for aadhi from 192.168.1.100 port 60211 ssh2
Feb 24 10:18:44 kali sudo: aadhi : TTY=pts/0 ; PWD=/home/aadhi ; USER=root ; COMMAND=/usr/bin/apt update
Feb 24 10:20:12 kali sshd[1102]: Failed password for root from 203.0.113.45 port 44211 ssh2
Feb 24 10:20:15 kali sshd[1102]: Failed password for root from 203.0.113.45 port 44211 ssh2
Feb 24 10:20:18 kali sshd[1102]: Failed password for root from 203.0.113.45 port 44211 ssh2
Feb 24 10:21:03 kali sshd[1120]: Accepted password for aadhi from 192.168.1.101 port 60333 ssh2
Feb 24 10:22:50 kali sshd[1155]: Failed password for invalid user test from 198.51.100.25 port 49822 ssh2
```

6. Suspicious IP Analysis

Authentication logs were reviewed to identify IP addresses associated with suspicious activity.

Several failed login attempts originated from the following IP addresses:

- 192.168.1.50
- 203.0.113.45
- 198.51.100.25

Analysis of Suspicious Activity

The IP address 192.168.1.50 repeatedly attempted authentication using the invalid username admin, indicating possible username enumeration or credential guessing attempts.

The IP address 203.0.113.45 repeatedly targeted the privileged root account and demonstrated multiple failed login attempts within a short time period.

Indicators of Suspicious Behavior

- Repeated failed authentication attempts
- Multiple attempts from the same source IP
- Attempts targeting privileged accounts
- Invalid username login attempts
- Rapid authentication failures

Such behavior commonly indicates:

- Brute-force attacks
- Credential guessing attempts
- Unauthorized access attempts
- Reconnaissance activity

Screenshot

```
Feb 24 10:20:12 kali sshd[1102]: Failed password for root from 203.0.113.45 port 44211 ssh2
Feb 24 10:20:15 kali sshd[1102]: Failed password for root from 203.0.113.45 port 44211 ssh2
Feb 24 10:20:18 kali sshd[1102]: Failed password for root from 203.0.113.45 port 44211 ssh2
```

7. Brute Force Detection Analysis

The authentication logs demonstrated repeated failed login attempts targeting the root account from the IP address 203.0.113.45.

The failed attempts occurred at:

- 10:20:12
- 10:20:15
- 10:20:18

The rapid repetition of authentication failures strongly suggests possible brute-force activity.

Indicators Observed

- Repeated failed login attempts
- Rapid authentication attempts within seconds
- Same source IP address repeatedly targeting the system

- Attempts against a privileged account (root)

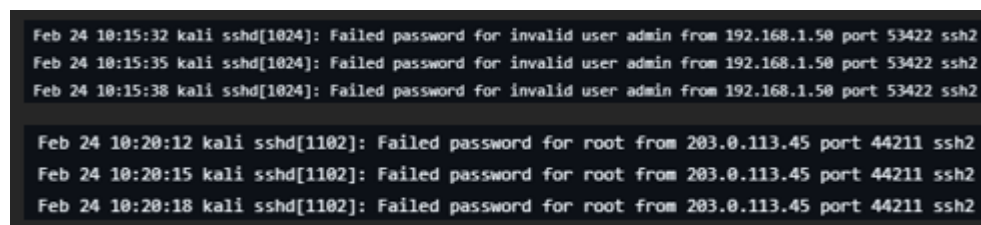
Security Perspective

Brute-force attacks commonly involve automated scripts or tools attempting repeated password guesses in order to gain unauthorized access.

Successful compromise of privileged accounts such as root may result in:

- Full system compromise
- Privilege escalation
- Unauthorized system modifications
- Data theft or malware installation

Screenshot



```
Feb 24 10:15:32 kali sshd[1024]: Failed password for invalid user admin from 192.168.1.50 port 53422 ssh2
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Feb 24 10:15:38 kali sshd[1024]: Failed password for invalid user admin from 192.168.1.50 port 53422 ssh2

Feb 24 10:20:12 kali sshd[1102]: Failed password for root from 203.0.113.45 port 44211 ssh2
Feb 24 10:20:15 kali sshd[1102]: Failed password for root from 203.0.113.45 port 44211 ssh2
Feb 24 10:20:18 kali sshd[1102]: Failed password for root from 203.0.113.45 port 44211 ssh2
```

8. Log Filtering & Investigation Techniques

Basic log filtering techniques were applied to isolate suspicious authentication events and improve investigation efficiency.

Logs were filtered based on:

- Failed password attempts
- Source IP addresses
- Privileged account targeting (root)
- Authentication-related events

Filtering suspicious entries helped identify:

- Repeated failed login attempts
- Brute-force behavior
- Suspicious source IP activity
- Privileged account targeting patterns

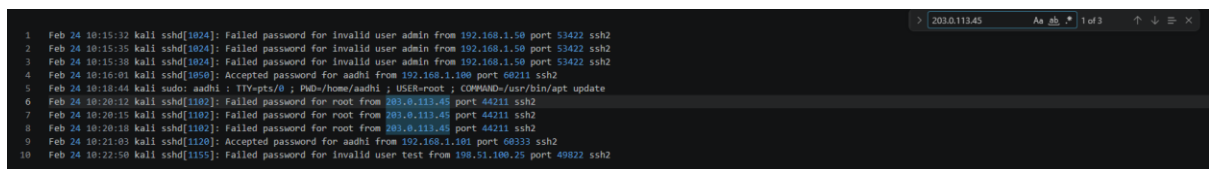
Investigation Workflow

The investigation process involved:

1. Identifying suspicious authentication events
2. Filtering repeated failed login attempts
3. Investigating suspicious source IP addresses
4. Correlating repeated authentication failures
5. Identifying possible brute-force patterns

These techniques simulate basic SOC analyst investigation workflows commonly used in security monitoring environments.

Screenshot



```
1 Feb 24 10:15:32 kali sshd[1024]: Failed password for invalid user admin from 192.168.1.50 port 53422 ssh2
2 Feb 24 10:15:35 kali sshd[1024]: Failed password for invalid user admin from 192.168.1.50 port 53422 ssh2
3 Feb 24 10:15:38 kali sshd[1024]: Failed password for invalid user admin from 192.168.1.50 port 53422 ssh2
4 Feb 24 10:15:40 kali sshd[1024]: Accepted password for aadh from 192.168.1.100 port 60211 ssh2
5 Feb 24 10:18:44 kali sudo: aadh : TTY=pts/0 ; PWD=/home/aadh ; USER=root ; COMMAND=/usr/bin/apt update
6 Feb 24 10:20:12 kali sshd[1102]: Failed password for root from 203.0.113.45 port 44211 ssh2
7 Feb 24 10:20:15 kali sshd[1102]: Failed password for root from 203.0.113.45 port 44211 ssh2
8 Feb 24 10:20:18 kali sshd[1102]: Failed password for root from 203.0.113.45 port 44211 ssh2
9 Feb 24 10:21:03 kali sshd[1120]: Accepted password for aadh from 192.168.1.101 port 60333 ssh2
10 Feb 24 10:22:50 kali sshd[1155]: Failed password for invalid user test from 190.51.100.25 port 49822 ssh2
```

9. Key Learnings

Through this project, the following concepts were understood:

- Structure and purpose of Linux authentication logs
- Failed login analysis techniques
- Detection of suspicious authentication behavior
- Identification of brute-force attack indicators
- Importance of privileged account monitoring
- Basic event correlation and log filtering techniques
- SOC-style security monitoring workflows
- Investigation of suspicious IP activity

10. Conclusion

This project provided practical exposure to SOC-focused log analysis and authentication monitoring using Linux authentication logs.

The analysis involved identifying failed login attempts, investigating suspicious IP addresses, detecting brute-force patterns, and applying basic log filtering techniques. Multiple indicators of suspicious authentication behavior were observed, including repeated failed logins targeting privileged accounts such as root.

Through this project, foundational cybersecurity concepts related to security monitoring, incident investigation, and threat detection were explored in a hands-on manner. The skills developed are directly relevant to SOC analyst and cybersecurity monitoring roles.

11. Repository Structure

```
soc-log-analysis/  
├── README.md  
├── logs/  
│   └── auth.log  
├── screenshots/  
│   ├── failed-logins.png  
│   ├── suspicious-ip-analysis.png  
│   ├── brute-force-pattern.png  
│   └── log-filtering.png  
└── report/  
    └── soc_log_analysis_report.pdf
```